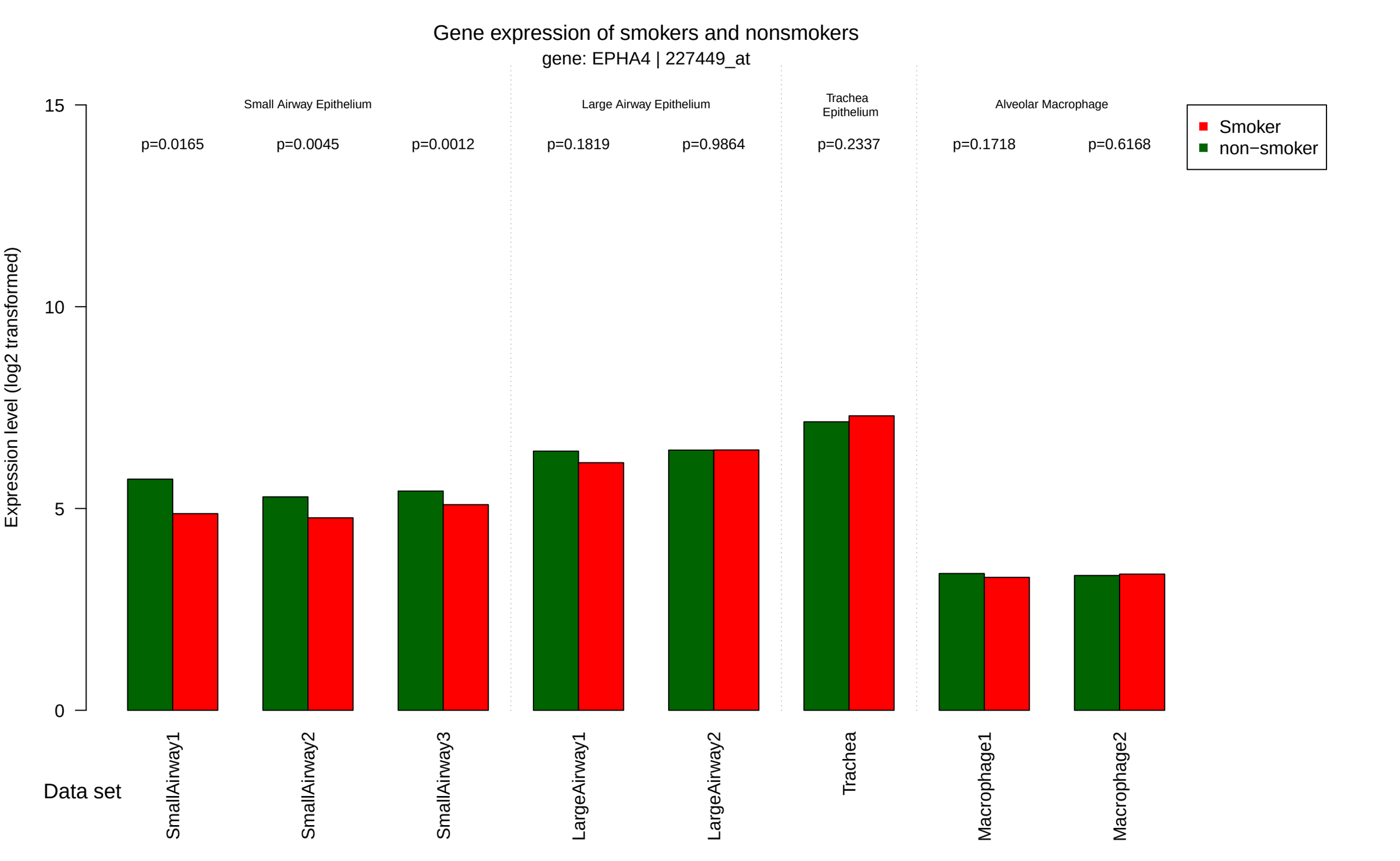
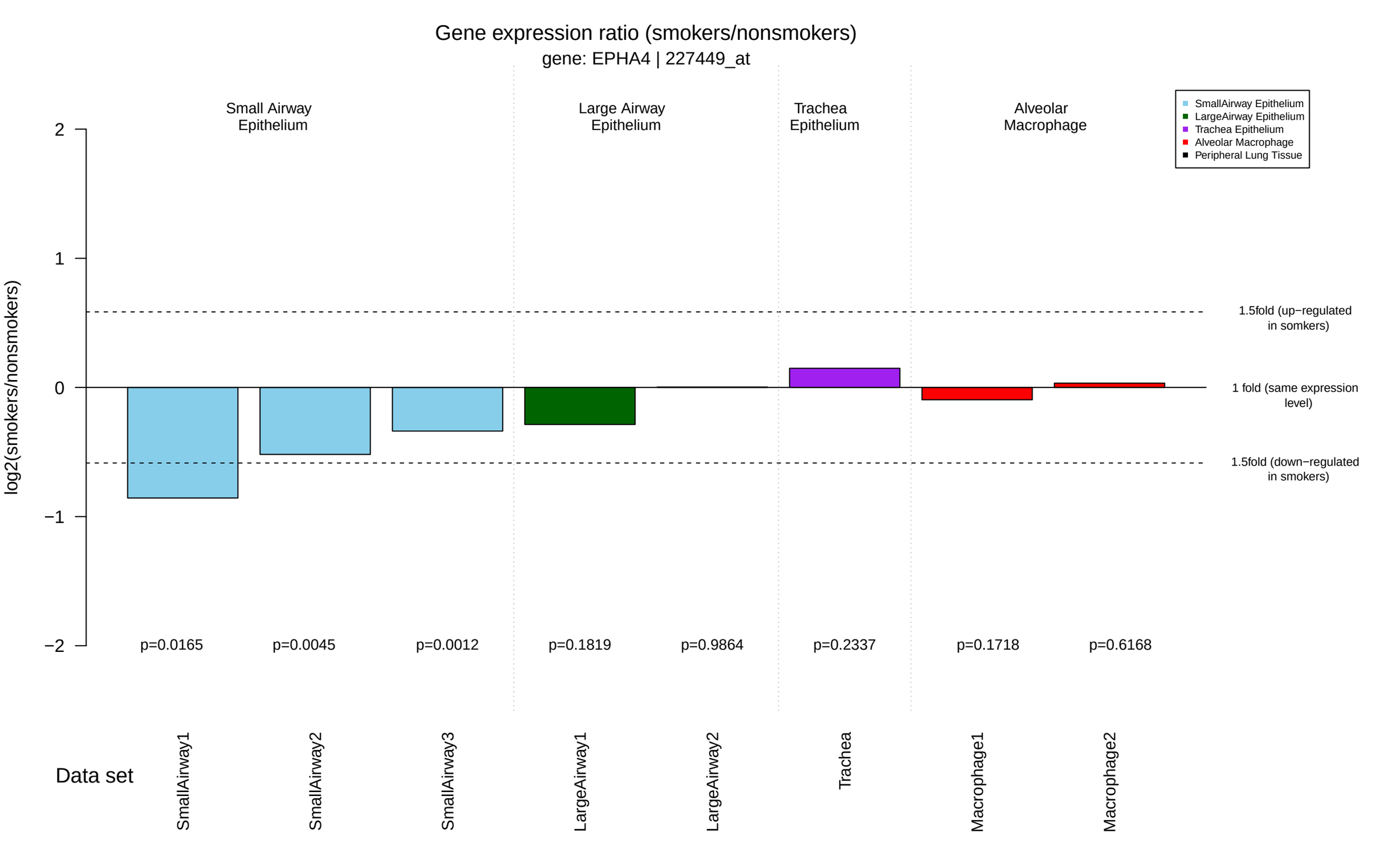
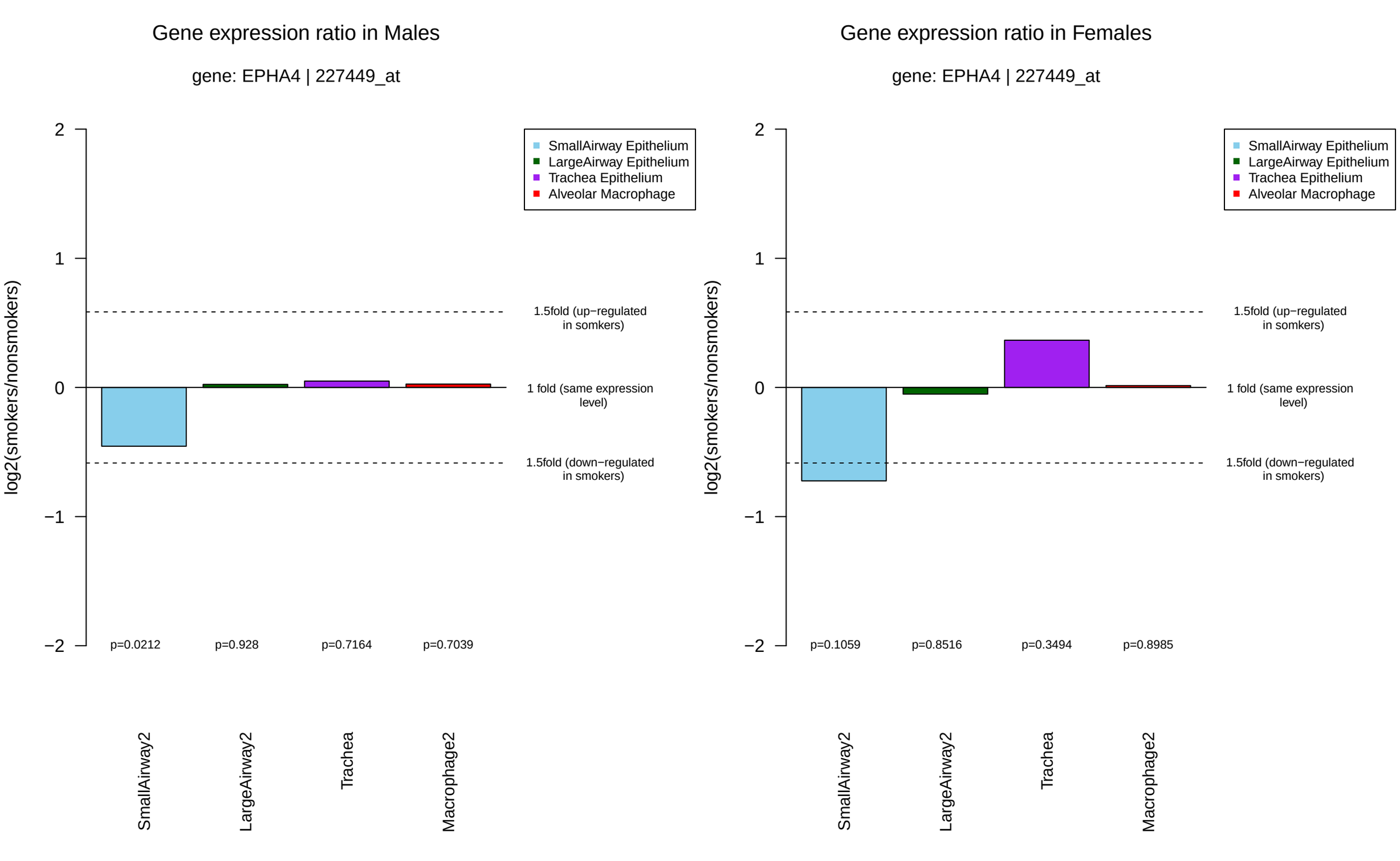
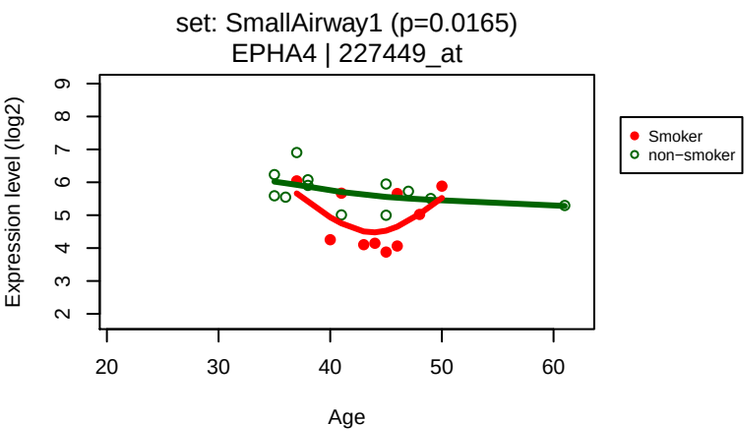
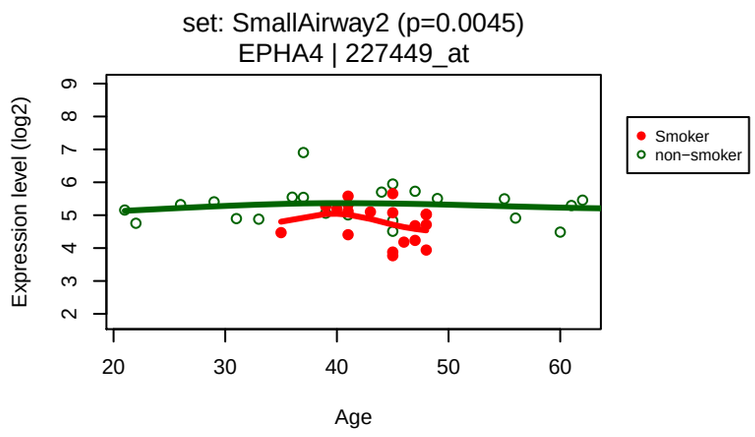
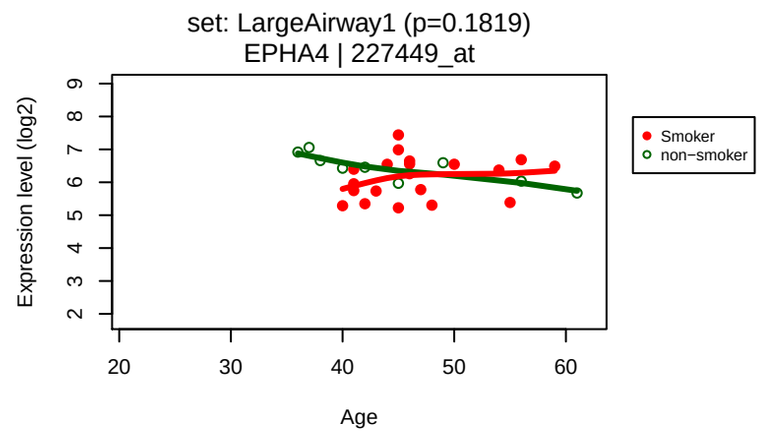

Gene expression of smokers and nonsmokers
gene: EPHA4 | 227449_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0165
p=0.0045
p=0.0012
p=0.1819
p=0.9864
p=0.2337
p=0.1718
p=0.6168
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

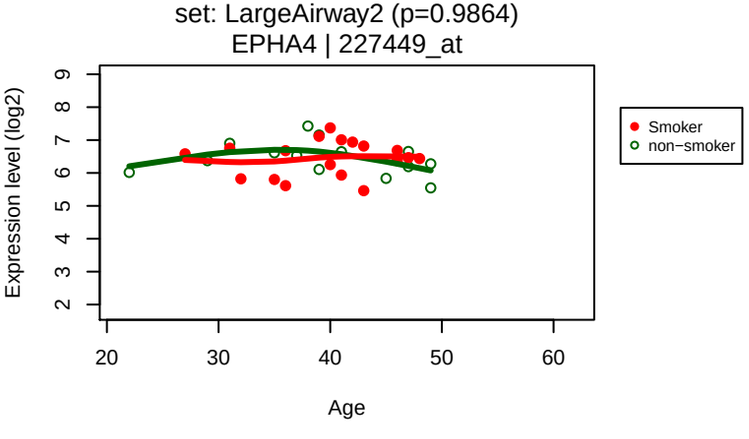

Gene expression ratio (smokers/nonsmokers)
gene: EPHA4 | 227449_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0165
p=0.0045
p=0.0012
p=0.1819
p=0.9864
p=0.2337
p=0.1718
p=0.6168
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

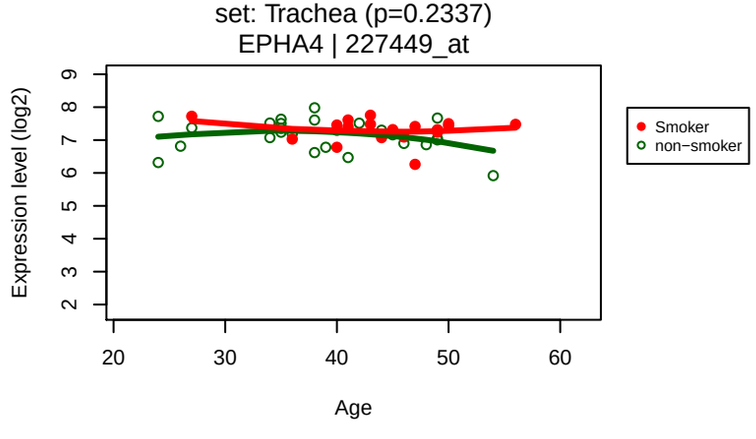

Gene expression ratio in Males
gene: EPHA4 | 227449_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0212
p=0.928
p=0.7164
p=0.7039
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: EPHA4 | 227449_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1059
p=0.8516
p=0.3494
p=0.8985
SmallAirway2
LargeAirway2
Trachea
Macrophage2

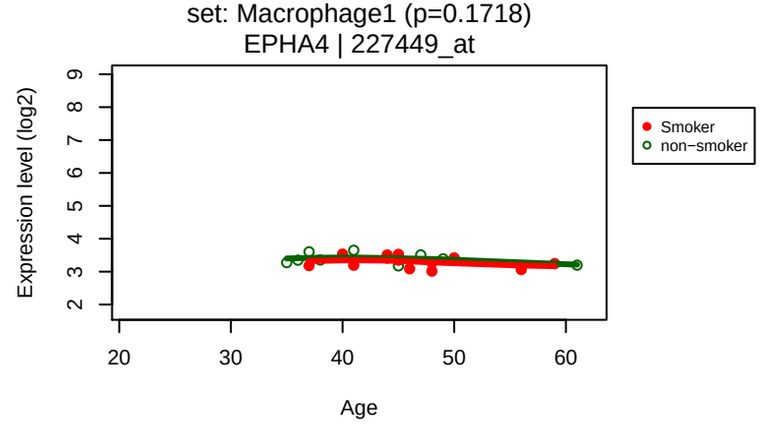

set: SmallAirway1 (p=0.0165)
EPHA4 | 227449_at
Expression level (log2)
Age
Smoker
non−smoker

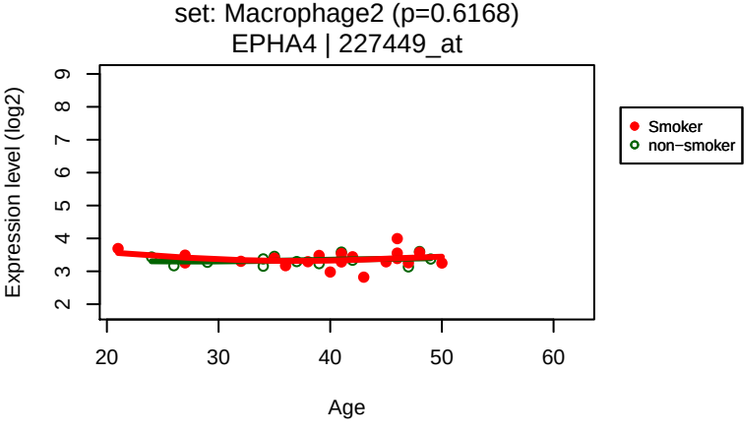

set: SmallAirway2 (p=0.0045)
EPHA4 | 227449_at
Expression level (log2)
Age
Smoker
non−smoker

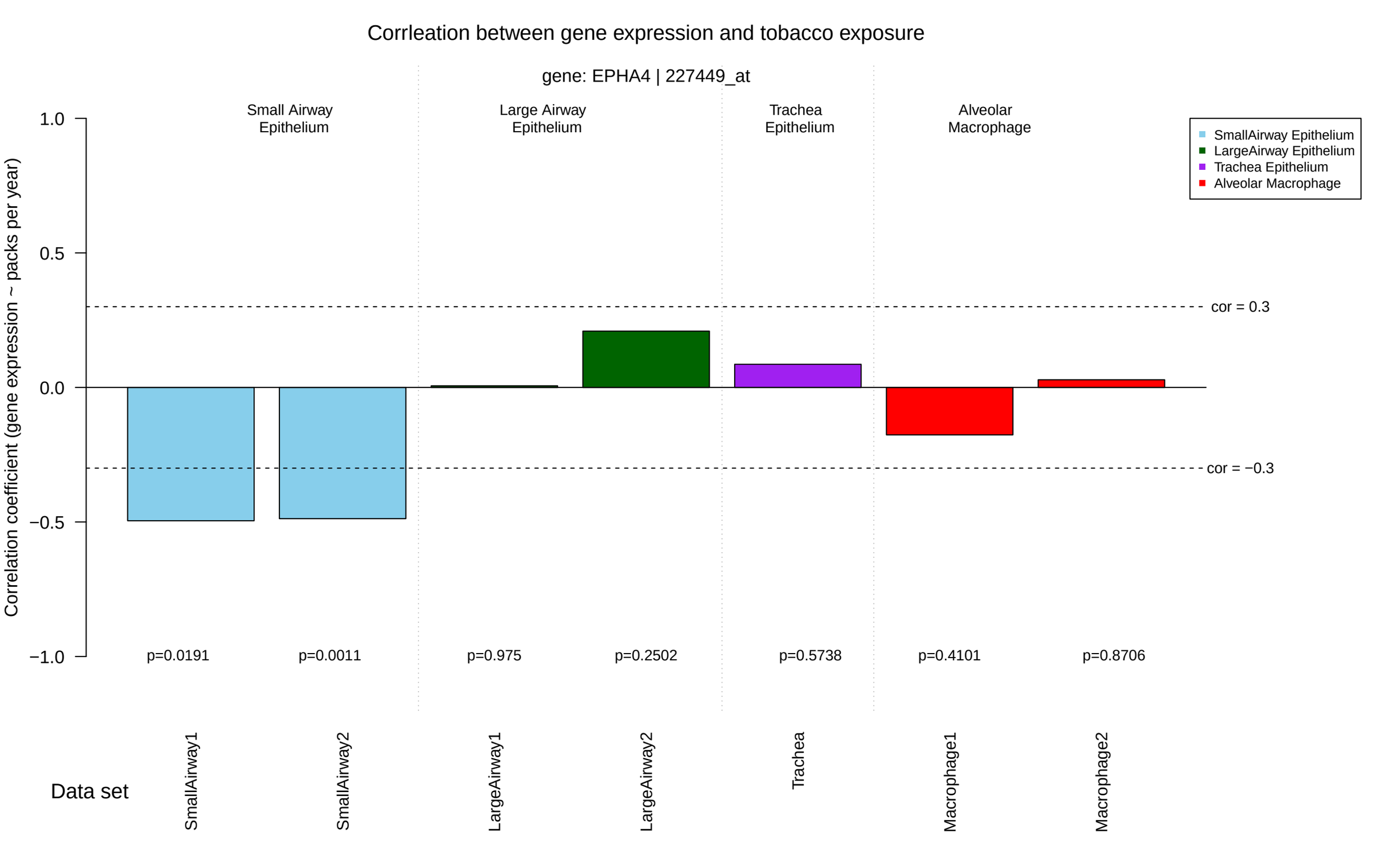

set: LargeAirway1 (p=0.1819)
EPHA4 | 227449_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9864)
EPHA4 | 227449_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2337)
EPHA4 | 227449_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1718)
EPHA4 | 227449_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6168)
EPHA4 | 227449_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: EPHA4 | 227449_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0191
p=0.0011
p=0.975
p=0.2502
p=0.5738
p=0.4101
p=0.8706
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2